Introduction to Economics

Class 7 — Elasticity

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Universitat de Barcelona · Group A6

15 October 2026

Quiz 4

10 minutes · pens down at 10:10

Circle the letter of your answer in the block at the top.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Quiz 4
10:10	Warm-up; Class 6, one more look
10:18	Olive oil abroad: a third less, the same money
10:21	Elasticity: the number, the midpoint
10:28	One line, many elasticities; in pairs
10:38	What makes demand elastic
10:43	Price $\times$ quantity: the box
10:50	Break
11:00	Topic fixed; Problem set 1 out
11:02	The olive oil answered; raise or cut?
11:14	Income and cross-price elasticities
11:27	Supply: when sellers can respond
11:38	In one line; before Monday 19 October

Warm-up: which curve moves, and which way?

Coach tickets, Barcelona–Valencia: for students, an inferior good

1. Students' incomes fall
2. Car-sharing rides get dearer
3. The coach fare falls
4. Diesel gets dearer

In pairs, two minutes; then hands per item.

Warm-up: which curve moves, and which way?

Coach tickets, Barcelona–Valencia: for students, an inferior good

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   demand shifts right: an inferior good
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   no shift: more tickets, along the demand curve
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   no shift: more tickets, along the demand curve
4. Diesel gets dearer
   supply shifts left; the fare rises along demand

Warm-up: what does the whole market buy?

Padel-court hours a month

price	Ana	Biel	the market
10 euros	5	3	
20 euros	3	1	

Carla joins: 2 hours at 10 euros, 1 at 20. The market curve?

In pairs, one minute; then hands.

Warm-up: what does the whole market buy?

Padel-court hours a month

price	Ana	Biel	the market
10 euros	5	3	$5 + 3 = 8$
20 euros	3	1	

Carla joins: 2 hours at 10 euros, 1 at 20. The market curve?

Warm-up: what does the whole market buy?

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10 euros	5	3	$\mathbf{5 + 3 = 8}$
20 euros	3	1	$\mathbf{3 + 1 = 4}$

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Warm-up: what does the whole market buy?

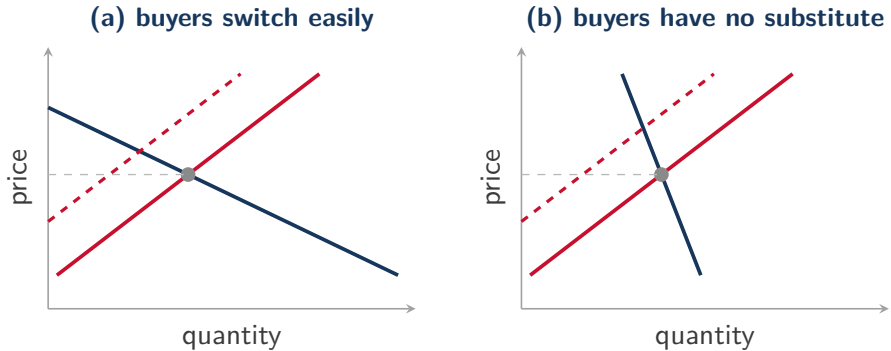
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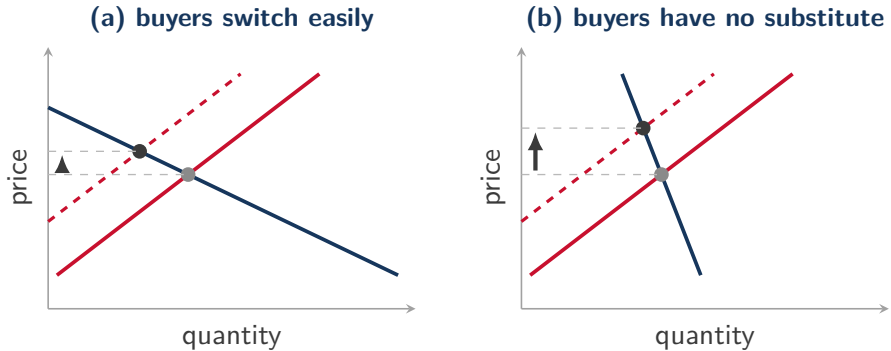
Carla joins: 2 hours at 10 euros, 1 at 20. The market curve?

shifts right: 10 hours at 10 euros, 5 at 20

Class 6 again: the same shift — where does the price jump more?



Class 6 again: the same shift — where does the price jump more?



(b). Same shift; the price jumps more where buyers cannot walk away.

Olive oil abroad in 2023: a third less oil, the same money

1.07 million tonnes → **0.68 million tonnes**

shipped abroad, 2022 → 2023

4.10 bn € → **4.15 bn €**

what came back, 2022 → 2023

What does that say about the buyers abroad?

Spain's olive-oil exports, customs data (AEAT), in MAPA's 2023 product report; 2023 provisional.

Elasticity: how much the quantity moves, in percentages

$$\text{price elasticity of demand} = \left| \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}} \right|$$

> 1 **elastic**: the quantity moves more than the price

< 1 **inelastic**: the quantity moves less than the price

$= 1$ **unit elastic**: the same

Percentages, so cups and tonnes compare. The absolute value drops the minus sign.

Two points, one number: percentages of the midpoint

The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day

computed from	price	cups	elasticity
the start, 2.00	+25 %	−25 %	1.0
the start, 2.50	−20 %	+33 %	1.67

Two points, one number: percentages of the midpoint

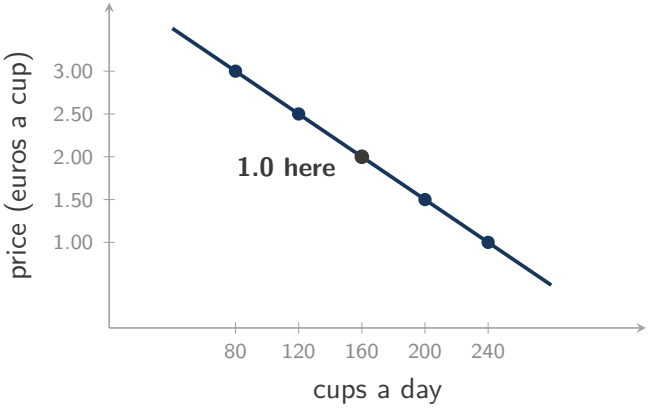
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computed from	price	cups	elasticity
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the start, 2.50	−20 %	+33 %	1.67
the midpoint	**+22 %**	**−29 %**	**1.29**

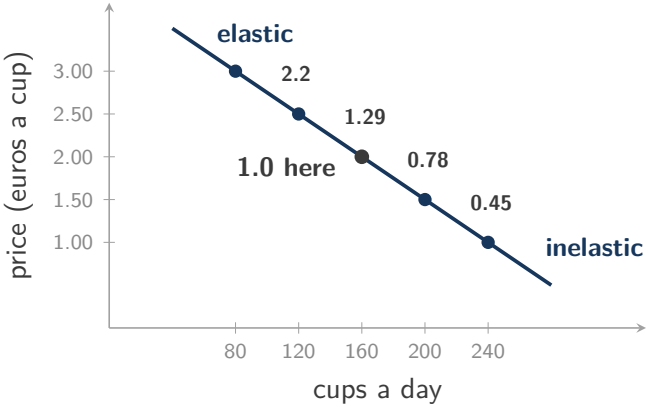
$$\left| \frac{(Q_2 - Q_1) \,/\, \frac{1}{2}(Q_1 + Q_2)}{(P_2 - P_1) \,/\, \frac{1}{2}(P_1 + P_2)} \right|$$

Divide each change by the average of the two values, not by the start.

One straight line: elastic at the top, inelastic at the bottom



One straight line: elastic at the top, inelastic at the bottom



Same slope everywhere; the percentages change:
few cups and a high price at the top, many cups and a low price at the bottom.

In pairs, four minutes: compute and classify

Elastic, inelastic, or unit elastic?

1. Bikinis at the bars: 5.00 $\rightarrow$ 5.50 euros, 200 $\rightarrow$ 150 a day

2. A single metro fare $+10\,\%$; rush-hour journeys $-2\,\%$

3. The zoo's ticket $+20\,\%$; visitors $-20\,\%$

Midpoint method where you have two points; then hands per item.

In pairs, four minutes: compute and classify

Elastic, inelastic, or unit elastic?

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   3.0, elastic
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Elastic, inelastic, or unit elastic?

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   3.0, elastic
2. A single metro fare $+10\,\%$; rush-hour journeys $-2\,\%$
   0.2, inelastic
3. The zoo's ticket $+20\,\%$; visitors $-20\,\%$

In pairs, four minutes: compute and classify

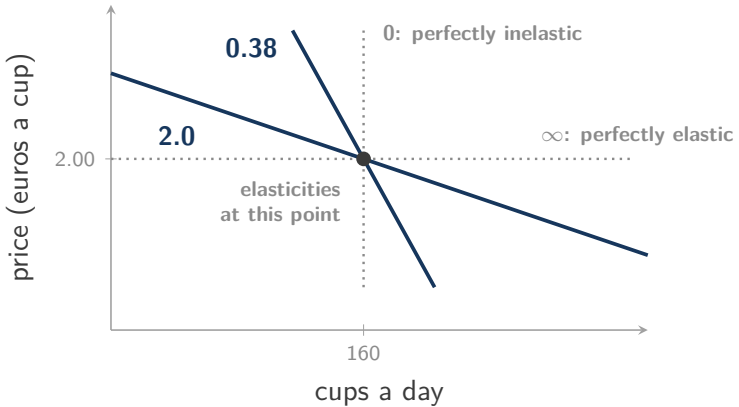
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   1.0, unit elastic

Demand is elastic when buyers can walk away

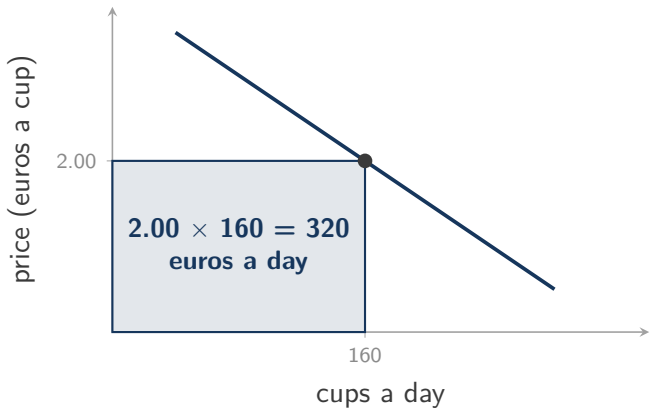
- Close substitutes nearby: Bar Nou, or the bar next door
- A luxury, not a necessity: a dinner out, not bread
- A narrow market: one bar's cup, not "coffee"
- Time to adjust: a year, not a week
- A big slice of the budget: rent, not chewing gum

Through one point: the flatter the line, the more elastic



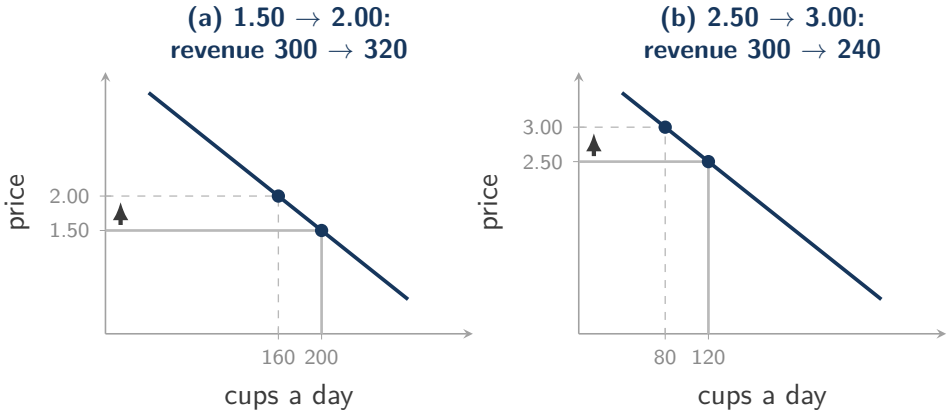
Same axes only. A line that looks flat on a stretched axis means nothing.

What buyers spend is what sellers get: price × quantity, a box

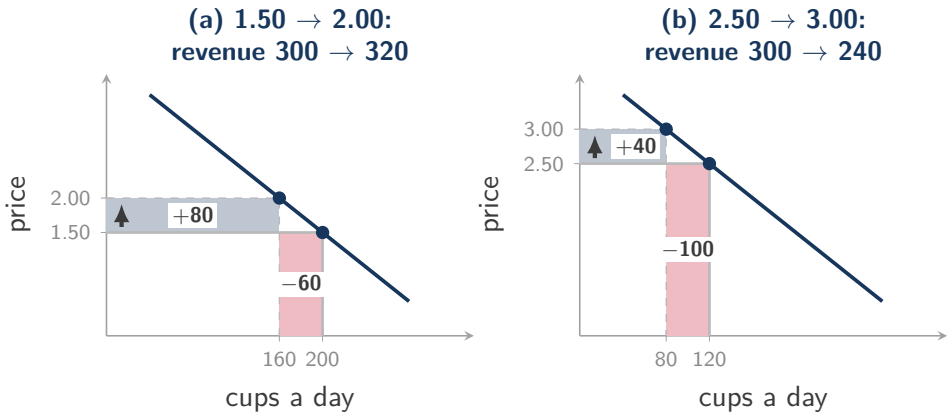


Total expenditure of buyers = **total revenue** of sellers: one box, two names.

A dearer cup: more money at the bottom of the line, less at the top



A dearer cup: more money at the bottom of the line, less at the top



Inelastic stretch (0.78): the gained strip wins. Elastic stretch (2.2): the lost one does.

Raise the price when demand is inelastic; cut it when elastic

demand	price up	price down
inelastic (< 1)	revenue up	revenue down
unit ($= 1$)	unchanged	unchanged
elastic (> 1)	revenue down	revenue up

Inelastic: revenue follows the price. Elastic: it runs the other way.

Which curve moves, and how many members? (optional)

Gym memberships in a city; price elasticity of demand 1.5

1. Memberships get 10 % dearer
2. Workout apps get dearer
3. Gyms' electricity bills double

In pairs, two minutes; then hands per item.

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   no shift: along demand, members −15 %
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   supply shifts left; price up, members down along demand

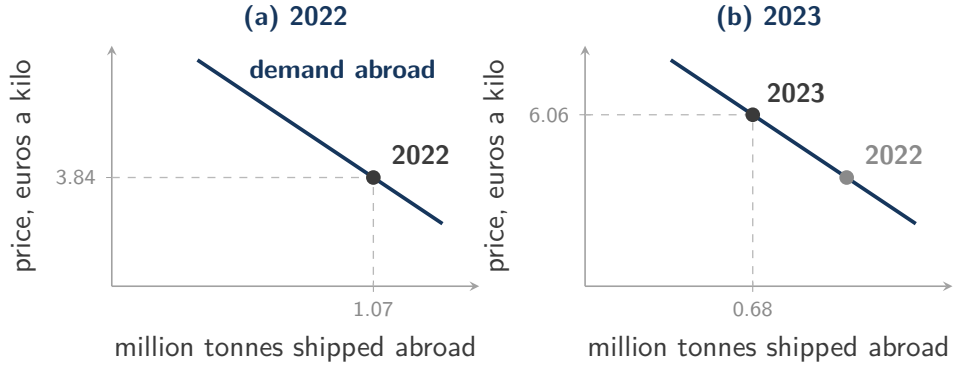
Break

Back at 11:00

Today: your topic is fixed; Problem set 1 is out

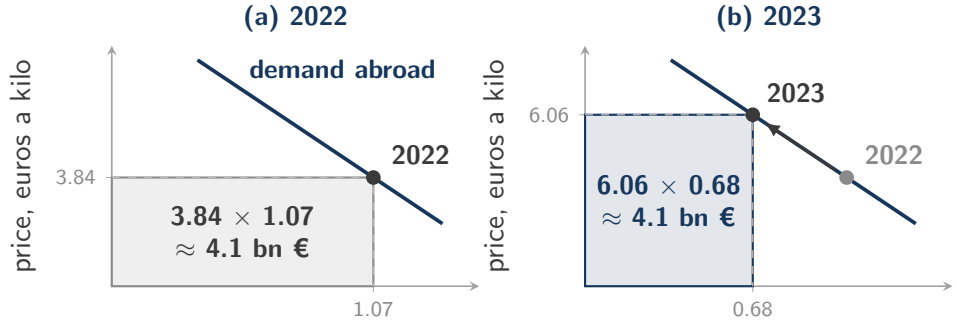
- Project: each group's question is settled today
- Not settled? See me after class
- Problem set 1 (Blocks 1–2): out today
- Due Thursday 22 October

Olive oil abroad: a third less, the same money — elasticity about 1



Spain's olive-oil exports 2022 and 2023: MAPA, Análisis del comercio exterior, aceite de oliva 2023 (customs data, AEAT; 2023 provisional). Average price = value ÷ tonnes. The curve is a sketch.

Olive oil abroad: a third less, the same money — elasticity about 1



Quantity −44 %, price +45 % (midpoint): elasticity 0.98.

Unit elastic: the price rise and the quantity fall cancel.

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In pairs: raise the price, or cut it?

More money, or less?

1. Rush-hour metro, elasticity 0.2: a dearer fare
2. Cinema on a Tuesday afternoon, 2.5: a cheaper ticket
3. Bar Nou alone among twelve bars, 4.0: a dearer cup
4. The neighbourhood's cups: 2.00 $\rightarrow$ 2.50, 160 $\rightarrow$ 120 a day

Three minutes; then hands per item.

In pairs: raise the price, or cut it?

More money, or less?

1. Rush-hour metro, elasticity 0.2: a dearer fare
   up: few leave
2. Cinema on a Tuesday afternoon, 2.5: a cheaper ticket
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   down: the next cup is next door
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   up: few leave
2. Cinema on a Tuesday afternoon, 2.5: a cheaper ticket
   up: many come
3. Bar Nou alone among twelve bars, 4.0: a dearer cup
   down: the next cup is next door
4. The neighbourhood's cups: 2.00 $\rightarrow$ 2.50, 160 $\rightarrow$ 120 a day
   revenue 320 $\rightarrow$ 300: down

Income elasticity: more income, more of most goods, less of some

$$\text{income elasticity of demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}$$

- **Normal good**: positive — a dinner out, a flat white
- **Inferior good**: negative — instant noodles, the cheapest own brand
- Among normal goods: luxuries above 1, necessities below 1

Class 5's 'income' shifter, now with a sign and a size.

Cross-price elasticity: substitutes positive, complements negative

$$\text{cross-price elasticity} = \frac{\%\text{ change in quantity of A}}{\%\text{ change in price of B}}$$

- The T-usual half price $\rightarrow$ fewer T-casual journeys: **substitutes**, positive
- Croissants dearer $\rightarrow$ fewer cafès amb llet: **complements**, negative

Zero: unrelated goods. Class 5's shift of demand, measured.

In pairs: which elasticity, what sign, what word?

Name it, compute it, give the word.

1. Income $+10\,\%$; instant noodles $-5\,\%$
2. Income $+10\,\%$; concert tickets $+15\,\%$
3. Tea $+20\,\%$; cups of coffee $+4\,\%$
4. Croissants $+10\,\%$; cafès amb llet $-3\,\%$
5. A cup $+10\,\%$; cups $-8\,\%$

Three minutes; then hands per item.

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   income, $\mathbf{1.5}$: **normal, a luxury**
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   cross-price, $\mathbf{+0.2}$: **substitutes**
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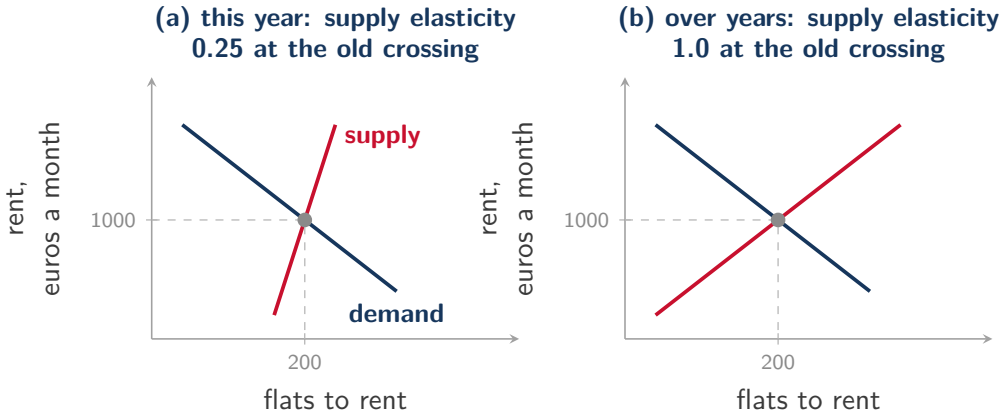
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   cross-price, $\mathbf{-0.3}$: **complements**
5. A cup $+10\,\%$; cups $-8\,\%$
   own price, 0.8: inelastic — not a cross-price

Supply elasticity: sellers respond when they can; time is the key

$$\text{price elasticity of supply} = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$$

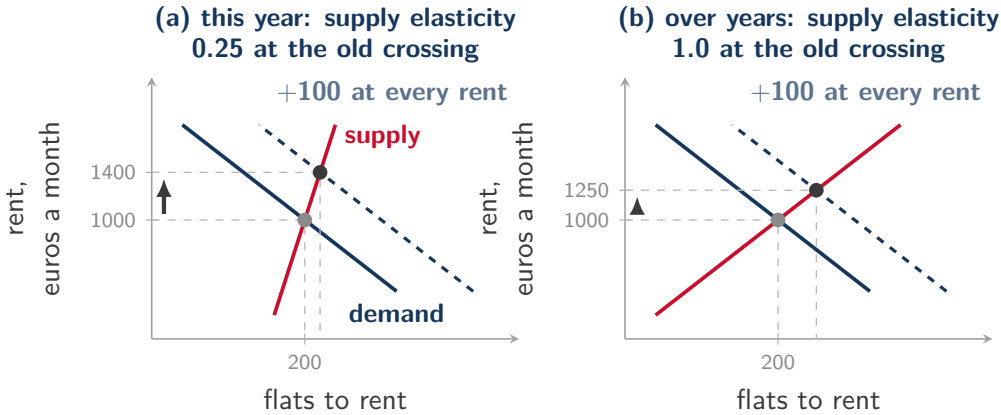
- This year's olives are already on the trees: near 0
- Spare capacity, stock in the warehouse: elastic
- Years to plant, build, hire: more elastic
- Vertical: perfectly inelastic — the seats in a hall tonight

Flats in Gràcia: more people want one; this year the rent jumps



Own numbers; not data.

Flats in Gràcia: more people want one; this year the rent jumps

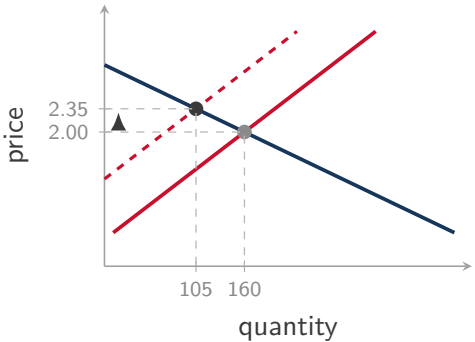


Same shift. Steep supply: rent +400, flats +20. Flatter: rent +250, flats +50.

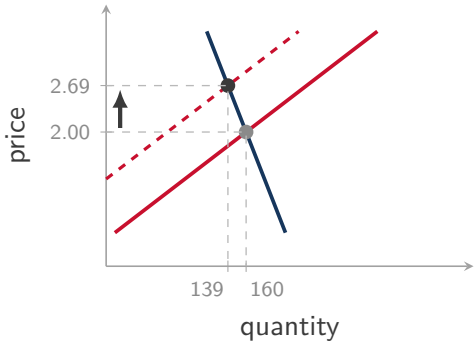
Own numbers; not data.

Price up 0.35 where buyers can switch, 0.69 where they cannot

(a) elasticity 2.0 at the old crossing: cups 160 → 105



(b) elasticity 0.38 at the old crossing: cups 160 → 139



Elastic buyers walk away: the quantity takes the hit. Inelastic buyers pay: the price does.

Two juice stalls: who should make what? (optional)

In one hour: Red stall, 6 smoothies or 3 sandwiches; Blue stall, 2 or 2

1. Red is faster at both. Can trade still help?
2. Who makes the sandwiches?
3. A sandwich price both accept?
4. Blue's hour becomes 2 smoothies or 1 sandwich. Now?

Two juice stalls: who should make what? (optional)

In one hour: Red stall, 6 smoothies or 3 sandwiches; Blue stall, 2 or 2

1. Red is faster at both. Can trade still help?
   yes: their opportunity costs differ
2. Who makes the sandwiches?
3. A sandwich price both accept?
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Two juice stalls: who should make what? (optional)

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   Blue: 1 smoothie each, against Red's 2
3. A sandwich price both accept?
   between 1 and 2 smoothies
4. Blue's hour becomes 2 smoothies or 1 sandwich. Now?
   no gain: now the same cost, 2 smoothies

What moves on the frontier? (optional)

A region grows almonds and makes bicycles

1. A better almond variety doubles the harvest
2. Laid-off bicycle workers get their jobs back
3. 5,000 workers move in, able to do either job

In pairs; draw it first.

What moves on the frontier? (optional)

A region grows almonds and makes bicycles

1. A better almond variety doubles the harvest
   only the almond end moves out
2. Laid-off bicycle workers get their jobs back
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What moves on the frontier? (optional)

A region grows almonds and makes bicycles

1. A better almond variety doubles the harvest
   only the almond end moves out
2. Laid-off bicycle workers get their jobs back
   from inside toward the frontier; the frontier stays
3. 5,000 workers move in, able to do either job

What moves on the frontier? (optional)

A region grows almonds and makes bicycles

1. A better almond variety doubles the harvest
   only the almond end moves out
2. Laid-off bicycle workers get their jobs back
   from inside toward the frontier; the frontier stays
3. 5,000 workers move in, able to do either job
   the whole frontier shifts out

Elasticity: the % change in quantity over the % change in price.

Supply shifts: demand inelastic, the price jumps; demand elastic, the quantity falls.

Revenue follows the price when demand is inelastic, runs against it when elastic.

Sellers: steep this year, flatter with time.

Before Monday 19 October

- Block 3 begins: surplus, efficiency, taxes and subsidies
- Read Mankiw & Taylor, Ch. 6, pp. 141–158 — consumer and producer surplus, market efficiency; Ch. 7, pp. 163–171 — taxes and subsidies
- Problem set 1 (Blocks 1–2): due Thursday 22 October
- Practise in Sextant: the units the course page lists

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